

HTML5 Exercises Answers

These answers are based on the HTML5 exercise sheet for the **Local Community Event Portal**, which asks for solutions covering base structure, navigation, styling, images, forms, events, media, storage, geolocation, and debugging tasks [file:1].

1. HTML5 Base Template

The base page should start with `<!DOCTYPE html>`, set the language with `<html lang=`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Local Community Event Portal</title>
</head>
<body>
  <!-- Navigation -->
  <nav>
    <a href="#home">Home</a>
    <a href="#events">Events</a>
    <a href="#contact">Contact</a>
  </nav>

  <!-- Main -->
  <main>
    <section id="home">
      <h1>Welcome to the Community Event Portal</h1>
    </section>
  </main>

  <!-- Footer -->
  <footer>
    <p>&copy; 2026 Local Council</p>
  </footer>
</body>
</html>
```

2. Navigation and Linking

The navigation must use a `<nav>` element with anchor links such as `<a href="#events`

```
<nav>
  <a href="#home">Home</a>
  <a href="#events">Events</a>
```

```

    <a href="#contact">Contact</a>
    <a href="help.html" target="_blank">Help</a>
</nav>

<section id="home">
    <h2>Home</h2>
</section>

<section id="events">
    <h2>Events</h2>
</section>

<section id="contact">
    <h2>Contact</h2>
</section>

```

3. Welcome Banner

This exercise requires a `div` with the `id` `welcomeBanner`, internal CSS for a blue background, an inline-styled `span` for a special offer, and a reusable `.highlight` class for emphasis [file:1]. The difference is that an `id` targets one unique element, while a `class` can be applied to many elements [file:1].

```

<head>
<style>
    #welcomeBanner {
        background-color: blue;
        color: white;
        padding: 15px;
        text-align: center;
    }

    .highlight {
        background-color: yellow;
        font-weight: bold;
    }
</style>
</head>
<body>
    <div id="welcomeBanner">
        Welcome back, resident!
        <span style="color:red; font-weight:bold;">Special Offer: Free entry today!</span>
    </div>

    <p class="highlight">Early registration is now open.</p>
</body>

```

4. Image Gallery

The gallery should use a table with 2 rows and 3 columns, where each cell contains an image, styling through a class for borders, and a caption describing the event shown [file:1]. Each image must include alt and title attributes for accessibility and better description [file:1].

```
<style>
    .event-img {
        width: 150px;
        height: 100px;
        border: 2px solid #333;
        padding: 4px;
    }
</style>

<table>
    <tr>
        <td>
            
            <p>Music Festival</p>
        </td>
        <td>
            
            <p>Food Fair</p>
        </td>
        <td>
            
            <p>Book Exhibition</p>
        </td>
    </tr>
    <tr>
        <td>
            
            <p>City Marathon</p>
        </td>
        <td>
            
            <p>Art Show</p>
        </td>
        <td>
            
            <p>Farmers Market</p>
        </td>
    </tr>
</table>
```

5. Event Registration Form

The form needs fields for name, email, date, event type, and message, along with placeholder, required, and autofocus attributes, plus an <output> element for confirmation after submission [file:1]. The task also specifically asks for CSS-based styling of the form layout [file:1].

```

<style>
  form {
    width: 400px;
    padding: 20px;
    border: 1px solid #ccc;
    border-radius: 8px;
    background: #f9f9f9;
  }
  input, select, textarea, button {
    width: 100%;
    margin-bottom: 10px;
    padding: 8px;
  }
</style>

<form onsubmit="document.getElementById('msg').value='Registration submitted success'">
  <input type="text" name="name" placeholder="Enter your name" required autofocus>
  <input type="email" name="email" placeholder="Enter your email" required>
  <input type="date" name="date" required>
  <select name="eventType" required>
    <option value="">Select event type</option>
    <option>Music</option>
    <option>Sports</option>
    <option>Workshop</option>
  </select>
  <textarea name="message" placeholder="Enter your message" required></textarea>
  <button type="submit">Register</button>
  <output id="msg"></output>
</form>

```

6. Event Handling

This section asks for `onblur` validation of a phone field, `onchange` to show a fee, `onclick` to confirm submission, `ondblclick` to enlarge an image, and keyboard event handling in a feedback textarea to count characters [file:1]. A single script can handle all of these user interactions clearly [file:1].

```

<input type="text" id="phone" placeholder="Enter phone number" onblur="validatePhone">
<select id="eventSelect" onchange="showFee()">
  <option value="">Choose event</option>
  <option value="100">Music - 100</option>
  <option value="200">Workshop - 200</option>
</select>
<p id="feeDisplay"></p>



<textarea id="feedback" onkeyup="countChars()" placeholder="Enter feedback"></textarea>
<p id="charCount">Characters: 0</p>

<button onclick="confirmSubmit()">Submit Feedback</button>

<script>

```

```

function validatePhone() {
    let phone = document.getElementById('phone').value;
    if (!/^d{10}$/.test(phone)) {
        alert('Enter a valid 10-digit phone number');
    }
}

function showFee() {
    let fee = document.getElementById('eventSelect').value;
    document.getElementById('feeDisplay').innerText = fee ? 'Selected event fee: ' +
}

function confirmSubmit() {
    alert('Feedback submitted successfully');
}

function enlargeImage() {
    document.getElementById('galleryImg').style.width = '300px';
}

function countChars() {
    let count = document.getElementById('feedback').value.length;
    document.getElementById('charCount').innerText = 'Characters: ' + count;
}
</script>

```

7. Video and Media Events

The video task requires a `<video>` element with controls and a source file, plus an `oncanplay` event that displays a message such as “Video ready to play” [file:1]. It also asks for `onbeforeunload` so users are warned before leaving an unfinished form page [file:1].

```

<video width="400" controls oncanplay="videoReady()">
    <source src="promo.mp4" type="video/mp4">
    Your browser does not support the video tag.
</video>
<p id="videoStatus"></p>

<script>
function videoReady() {
    document.getElementById('videoStatus').innerText = 'Video ready to play';
}

window.onbeforeunload = function() {
    return 'Your form is not finished. Are you sure you want to leave?';
};
</script>

```

8. Saving Preferences

The sheet asks for saving the selected event type in `localStorage`, restoring it on reload, and clearing both `localStorage` and `sessionStorage` using a button [file:1]. This allows the portal to remember user preferences for returning visitors [file:1].

```
<select id="preferredEvent" onchange="savePreference()">
  <option value="">Choose preferred event</option>
  <option value="Music">Music</option>
  <option value="Sports">Sports</option>
  <option value="Workshop">Workshop</option>
</select>

<button onclick="clearPreferences()">Clear Preferences</button>

<script>
window.onload = function() {
  let saved = localStorage.getItem('preferredEvent');
  if (saved) {
    document.getElementById('preferredEvent').value = saved;
  }
};

function savePreference() {
  let selected = document.getElementById('preferredEvent').value;
  localStorage.setItem('preferredEvent', selected);
}

function clearPreferences() {
  localStorage.clear();
  sessionStorage.clear();
  alert('Preferences cleared');
}
</script>
```

9. Geolocation

The geolocation question requires a button that finds nearby events by calling `navigator.geolocation.getCurrentPosition`, shows latitude and longitude, handles errors such as denial or timeout, and uses high-accuracy options [file:1]. This satisfies the practical mapping requirement in the exercise [file:1].

```
<button onclick="findLocation()">Find Nearby Events</button>
<p id="locationResult"></p>

<script>
function findLocation() {
  if (navigator.geolocation) {
    navigator.geolocation.getCurrentPosition(showPosition, showError, {
      enableHighAccuracy: true,
      timeout: 10000,
      maximumAge: 0
    });
  }
};
```

```

    } else {
        document.getElementById('locationResult').innerText = 'Geolocation is not supported';
    }
}

function showPosition(position) {
    document.getElementById('locationResult').innerText =
        'Latitude: ' + position.coords.latitude + ', Longitude: ' + position.coords.longitude;
}

function showError(error) {
    let msg = '';
    switch(error.code) {
        case error.PERMISSION_DENIED:
            msg = 'User denied the request for Geolocation.';
            break;
        case error.POSITION_UNAVAILABLE:
            msg = 'Location information is unavailable.';
            break;
        case error.TIMEOUT:
            msg = 'The request to get user location timed out.';
            break;
        default:
            msg = 'An unknown error occurred.';
    }
    document.getElementById('locationResult').innerText = msg;
}
</script>

```

10. Debugging with Chrome DevTools

The final task asks students to use Chrome DevTools to inspect elements, test style changes, read JavaScript logs in the Console, and add breakpoints to observe variable values while reloading the page [file:1]. In VS Code, opening the HTML file, checking the script section, and tracing event functions makes debugging easier before testing again in the browser [file:1].

- Use **Inspect Element** to check layout and edit CSS temporarily [file:1].
- Use the **Console** tab to view `console.log()` output and JavaScript errors [file:1].
- Use **Sources** in DevTools to place breakpoints and monitor script execution [file:1].
- Reload the page after changes to test whether the issue is resolved [file:1].

Complete Note

These answers provide a ready-to-write solution set for all 10 HTML5 exercises in the attached module sheet, using the required tags, form controls, events, storage methods, geolocation handling, and browser debugging steps described in the PDF [file:1].